

Shakhawat Hosen Tanim, Ph.D.

Assistant Professor, Landscape Architecture and Regional & Community Planning

Faculty Scholar, Kansas Water Institute

Kansas State Universitystanim@ksu.edu / [Website](#) / [Google Scholar](#)**Research Interests**

Human–environment interactions

Geospatial data science and GeoAI

Flood and storm risk assessment and decision-making

Infrastructure interdependencies

Emergency and health response

Education

- 2023 University of South Florida, Tampa, Florida
Ph.D. in Geography, Environmental Science & Policy.
Dissertation: Who Goes and Who Stays: Predicting Hurricane Evacuation for Local Neighborhoods.
- 2021 University of South Florida, Tampa, Florida
Graduate Certificate in Geographic Information Systems.
- 2013 University of Dhaka, Bangladesh
Master of Science in Disaster Management.
Thesis: Vulnerability and Preparedness of Hospitals for a Potential Earthquake.
- 2011 Bangladesh University of Engineering and Technology, Bangladesh
Bachelor of Urban and Regional Planning.
Thesis: Modeling Driving Behavior of CNG- and Gasoline-Driven Vehicles.

Professional Experience

- 2026- present Kansas State University, *Assistant Professor*
- 2023- 2026 Clemson University, *Postdoctoral Fellow*
- 2021-2023 Kimley-Horn, *Geospatial Scientist*
- 2015-2021 University of South Florida, *Ph.D. Researcher & Graduate Teaching Assistant*
- 2017-2019 Center for Urban Transportation Research (USF), *Research Assistant*
- 2011-2015 Sheltech (Pvt.) Ltd., *Assistant Urban Planner*

Grant Experience

- 2026 *Data-Driven Approaches for Opioid Use Disorder Treatment, Recovery, and Overdose Prevention in Rural Communities via Mobile Health Clinics* (National Institutes of Health (NIH); **Awarded, Co-investigator**).
- 2025 *Increasing Mobile Health Clinic Utilization in Rural Communities* (South Carolina Rural Primary Healthcare; **Awarded, Co-investigator**).
- 2023 *Dam Sector Regional Risk Resilience Program (DSR3P), Year 2* (U.S. Army Corps of Engineers; Not awarded, **Co-investigator**).
- 2022 *Trustworthy, Explainable, and Intelligent Digital Twin Framework for Geospatial Analytics* (Taylor Geospatial Institute; Not awarded, **Co-investigator**).

Key Publications

Under review

- 2026 Woo, M., **Tanim**, S. H., & Rennert, L. Moral bounded rationality and the reconstruction of structural silence through compassionate AI. *Information Systems Research* (under review).
- 2026 **Tanim**, S. H., Murray-Tuite, P., & Deegan, M. Mapping varied functions of hydropower dam and reservoir systems for resilience. *Resilient Cities and Structures* (submitted).
- 2026 Tabassum, N., **Tanim**, S. H., Bramlett, K., Gezer, F., Rennert, L., & Nolan, M. (2025). Seasonality, environmental drivers, and predictive modeling of tick abundance: An integrated analysis. *Parasites & Vectors* (submitted).

Published

- 2026 **Tanim**, S. H., White, D., Witrick, B., & Rennert, L. Optimizing mobile health clinic placement via geospatial modeling. *Public Health in Practice*, 100805.
- 2026 Kabir, M. E., **Tanim**, S. H., Sellnow, D. D., Luteria, G. L. P., & Rennert, L. (2026). Teaching AI to Decode Vaccine Hesitancy Narratives: A Few-Shot Learning and Topic Modeling Approach. *Big Data and Cognitive Computing*, 10(5), 159.
- 2025 Hogewood, L., **Tanim**, S. H., Deegan, M., & Ladner, D. Resilience assessment matrix development for community water supply. *Natural Hazards Review*, 26(6), 101061.
- 2025 Wu, J., **Tanim**, S. H., & Rennert, L. A deep learning approach for enhancing pandemic prediction: A retrospective evaluation of transformer neural networks and multi-source data fusion for infectious disease forecasting. *Epidemics*, 53(2025), 100865
- 2024 **Tanim**, S., Woo, M., & Rennert, L. A machine learning and GIS-based approach for mobile health clinic placement: Identifying factors for successful deployments in South Carolina. *Proceedings of the 2024 IEEE International Conference on Big Data*, Washington, DC.
- 2024 Bhatta, N., **Tanim**, S. H., & Murray-Tuite, P. Dynamics of link importance through normal conditions, flood response, and recovery. *Sustainability*, 16(2), 819.
- 2023 **Tanim**, S. H., Reader, S., & Hu, Y. Predicting hurricane evacuation for local neighborhoods across a metropolitan region. *International Journal of Disaster Risk Reduction*, 99, 104117.
- 2022 **Tanim**, S. H., Wiernik, B. M., Reader, S., & Hu, Y. Predictors of hurricane evacuation decisions: A meta-analysis. *Journal of Environmental Psychology*, 79, 101742.
- 2020 Yu, Q., & **Tanim**, S. H. A preliminary exploration of the cooling effect of tree shade in urban landscapes. *International Journal of Applied Earth Observation and Geoinformation*, 92, 102161.
- 2019 Acheampong, M., **Tanim**, S. H., Ananga, E. Is Ghana ready to attain SDG 7: A comprehensive assessment of its renewable energy potential and pitfalls. *Energies*, 12(3).
- 2017 **Tanim**, S. H., & Tobin, G. A. Social factors and evacuation vulnerability: An application in Pinellas County, Florida. *Papers in Applied Geography*, 4(2), 123-136.

Manuscripts in Progress

- Development of an interactive web mapping tool for geospatial decision support system for health
- Developing a method to predict fuel consumption for evacuating vehicles
- Patient use of mobile health services through geospatial modeling of access and environment

Select Presentation and Conferences

Seminar Talk

- 2026 Geospatial Analytics for Public Health: Experience from the Center for Public Health Modeling and Response. Talk at Clemson Center for Geospatial Technologies in Clemson University.
- 2026 Decision-Ready GIScience: Linking Systems, People, and Place. Talk at School of Geosciences in University of South Florida.
- 2025 Risk, Resilience, and Response: Decision-Ready Analytics. Talk at Institute for Risk Management and Insurance Innovation (IRMII) in University of North Carolina at Chapel Hill.
- 2023 Towards a Resilient and Sustainable Future: Innovative Approaches in Environmental Sciences. Talk at Department of Earth, Environmental, and Atmospheric Sciences in Western Kentucky University.

Conferences

- 2026 Interactive Web Tool to Optimize Mobile Health Clinic Placement via Geospatial Modeling. CSTE 2026 Annual Conference, EpiVision Session, Boston, MA.
- 2024 Strategic Placement of Mobile Health Clinics: Integrating Geospatial Machine Learning and Location-Allocation. American Geophysical Union (AGU) Fall Meeting, Washington, DC.
- 2024 A Geospatial Machine Learning Approach to Explore Factors Driving Mobile Health Utilization. IEEE Workshop on Trustworthy Machine Learning for Healthcare, Washington, DC.
- 2021 Effect of Predictor Heterogeneity on Evacuation-Demand Models. AAG Annual Meeting.
- 2020 A Statistical Meta-Analysis of Hurricane Evacuation Decision Models. Southeastern Division of the American Association of Geographers (SEDAAG).
- 2020 Modeling the Geospatial Demographics of Hurricane Evacuation for Tampa Bay, Florida. USF School of Geosciences Colloquium, Tampa, FL.
- 2017 Modeling Vehicle Conversion to CNG-Driven Vehicles Using Mixed Logit and Latent Class Logit: A Case Study in Dhaka, Bangladesh. University Transportation Centers (UTC) Conference, Gainesville, FL.
- 2016 Integrating Social and Transportation Aspects of Evacuation with GIS. SEDAAG, Columbia, SC.
- 2016 Documenting Social Factors and Evacuation Vulnerability: Pinellas County, Florida. Applied Geography Conference, Louisville, KY.
- 2016 Understanding How Urban Landscape Heterogeneity and Socioeconomic Structure Affect Urban Heat-Island Effects in Tampa. Research Symposium: Celebrating Geographies of Space, Tampa.

Teaching Experience

Instructor

- 2026-27 K-State Department of Landscape Architecture and Regional & Community Planning (LARCP)
 - CDPLN 712: Sustainable Communities
 - LAR 420: Socio-Ecological Systems
- 2018-21 USF School of Geoscience
 - GIS-4043, GIS-5049: Geographic Information System
 - EVR-4027: Wetland Environment
- 2018 USF College of Engineering
 - CGN-4933: GIS for Transportation (guest lecture)

Student Supervising

- 2023- Navin Bhatta (committee member), MS Student, Clemson University.
2024 *Thesis: Dynamics of Link Importance Across Flood Disaster Phases and Flood Impacts*

Awards

- 2021 Jeanne X. Kasperson Award by Hazards, Risks, and Disasters Specialty Group, AAG.
2021 Dissertation Completion Fellowship by USF Office of Graduate Studies.
2016-2019 Tharp Research Fellowships by USF School of Geoscience.

Media Coverage

- 2019 [University of South Florida Podcast- Calling: Earth](#), on “Behavioral modeling for transportation and hurricane evacuation.” September 2019.

Professional Memberships

American Geophysical Union | American Association of Geographers | Institute of Transportation Engineers | Council of State and Territorial Epidemiologists | University Consortium for Geographic Information Science | MultiSector Dynamics | Society for Risk Analysis

Service Committee Member

Clemson Faculty GIS Advisory Group

Methods and Computational Tools

Methods: Geospatial data science; GIS and spatial modeling; machine learning and deep learning; network analysis; optimization and location-allocation; behavioral modeling; agent-based simulation; forecasting; decision-support systems.

Tools: Python, R, ArcGIS Pro/Online, QGIS, PostGIS, Google Earth Engine, PyTorch, TensorFlow, scikit-learn, HEC-RAS/HMS, Vensim, NetLogo, Leaflet, Folium, and Streamlit.